

Course Outline

National University of Computer & Emerging Sciences
Lahore Campus



Course:	Object Oriented Programming		
Course code:	CS2012	Year/semester:	Fall 2025
Program:	BFT-3A, BFT-3B	Units/Cr Hrs:	3 Credit hour
Department:	Fast School of Management	Instructor:	
Course type:	Core Course	Email:	
Pre-requisite(s):	Programming for Business	Phone:	
Prepared by:		Consultation hours:	To be announced in class
Approved by:		Approval date:	

COURSE DESCRIPTION

This introductory course on object-oriented programming is designed to develop an understanding of the fundamental concepts of object-oriented programming, design patterns, best practices, principles, and conventions. The course covers basic to advanced-level concepts of objects, classes, inheritance, abstraction, polymorphism, basic design patterns, SOLID principles, UML diagrams, etc. The aim is to design the solution to utilize the concepts of object-oriented programming and design principles.

COURSE OBJECTIVES

Upon successful completion of the lab students should be able to:	
1	Understand the underlying knowledge of object-oriented programming
2	Develop an understanding of object-oriented design artifacts and their mapping to object-oriented programming.
3	Understanding the basic design patterns and programming principles.
4	Design and implement object-oriented programming knowledge to design and develop a system.

COURSE CONTENT

LO#	Learning Outcome Statement	PO	BT Level
LO1	Introduction, classes, objects, constructors, functions, conditional statement, loops, diagrams, flowcharts	PLO 1,2	Level1 Understand
LO2	Understanding the fundamentals of Inheritance, polymorphism, abstraction, data hiding, class members accessors	PLO2, 3,4	Level 2,3 Understand and application
LO3	Learning the concept of design patterns, principles, and programming practices to utilize the concepts of object-oriented programming.	PLO5	Level 2,3 Apply
LO4	Understanding and applying Object-oriented programming principles, design patterns for different use cases, and documenting them.	PLO 7	Level 3: Apply

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LAB CONTENTS

Session		Topic
LO1	1	Introduction to the course: ● Grading, project outline, and brief introduction about object-oriented programming Basic concepts of programming (Recap): ● Data types, arrays, strings, conditional statements, loops, operators, lists, data frames, functions. Problem Solving: ● Way to approach the problem ● Analysis ● Develop Algorithm ● Divide and conquer rule ● Real-world scenarios and problem-solving
LO1	2	Classes and objects: ● Introduction to Classes ● Object ● Convention of classes ● Attributes and methods ● Class diagram ● Self keyword ● Variable scope ● Real-world scenarios and problem-solving Problem Solving: ● Approaching a problem with classes
LO1	3	Inheritance: ● Types of relations ● Psuedo code ● Flow charts ● UML Diagrams ● Constructors ● Destructors ● Multiple Inheritance ● Diamond problem and overcoming this problem ● Real-world scenario Problem Solving: ● Approaching a problem with inheritance

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LO2	4	Encapsulation: • Encapsulation • Public, static, protected, and private class members • Controlled access • Data hiding, getters, setters Project Ideas submission: • Presentation of project ideas • Approval from the instructor • Submit a project proposal document containing diagrams Problem Solving: • Approaching a problem with encapsulation
LO2	5	Polymorphism: • Introduction • Overriding and overloading • Real-world scenario Finalization of project ideas • The project ideas must be finalized. Problem Solving: • Approaching a problem with Polymorphism
LO2	6	Interfacing • Introduction • Real-world problem File Handling • CRUD operation using files • Streams, stream object, character I/O, binary I/O • Readers and writers • Multiple objects, positions, offsets Problem Solving: • Approaching a problem with Interfacing
		Mid Term

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LO3	8	Design patterns • Singleton design pattern Object-oriented design • Forward engineering of the class diagram to code • Reverse Engineering of code to class diagram
LO3	9	Project progress and presentation • Progress checkpoint with evaluation
LO3	10	Creational Design patterns (Basic level) • MVC design pattern • Abstract, Builder, Factory, Prototype, Singleton Structural Design patterns (Basic level) • Adapter, Bridge, Composite, Decorator, Behavioral Design Patterns (Basic level) • Observer, Command, Chain of responsibility, iterator,
LO3	11	Introduction to JSON/ XML • Introduction Principles • SOLID ○ Single Repository Principal (SRP) ○ Open close Principal (OCP) ○ Liskpv Substitution Principal (LSP) ○ Interface Segregation Principle (ISP) ○ Dependency Inversion Principal (DIP) • DRY (Dont Repeat Yourself) • YAGNI (You Aint Gonna Need It) • COC (Convention over Configuration) • LOD (Law of Demeter)
LO3	12	GUI (Graphical User Interface)
LO4	13	Basic Software Engineering • Introduction • Software Development Life Cycle (SDLC) • Development Models (Waterfall, Agile) • Project Management Process • Cyclomatic complexities
LO54	14	System Design • Designing a complex system as a class activity using OOP principles Final presentations of the project (2nd half) - Presentation with viva

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LO4	15	Practice with a case study (1st half) - Solving a real-world case study that will use all the principles that have been studied above. Final presentations of the project (2nd half) - Presentation with viva
LO4	16	Practice and case studies ● Business case studies and coding practice in the second half

Marks distribution

Grading mechanism	Weight
Announced/surprise quizzes	10%
Assignments	10%
Class participation (Daily class activities participations)	5%
Project with documentation (Project 10%, documentation 5%)	15%
Midterm exam	20%
Final exam/ final project	40%

Textbook& reference material

Textbook:

- Starting out with Python" by Tonny Gaddis

Administrative instructions

Grading Criteria:

Grading will be relative/absolute.

Missing exams or quizzes

- *There will be no retake of missed exams or quizzes unless approved or agreed before.*
- Quizzes, mid-term, and final exams will be conducted in the class. Make sure that you come prepared.

Class Policies

- Please note that any exception for one student is unfair to all other students, so don't expect any.
- Talking during a lecture is not permitted. It is disrespectful and disruptive to other class members and the instructor.

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- Completion of the daily lab tasks will strictly be considered as important and not being able to comply will result in the loss of marks in the class participation.
- If a student is on leave, they will not receive compensation for missed class participation marks. The course includes 14 sessions, with 1 absolute mark allocated for each lab participation. The maximum achievable mark for class participation is 5. Consequently, if a student misses a lab, they are required to make up for it in the next available lab session. No retakes for missed labs will be permitted.

Attendance policy

As per FAST NUCES attendance policy

Scholastic dishonesty

In the lectures and all other academic activities, students are expected to uphold the highest standards of academic integrity. Any form of scholastic dishonesty is an affront to the pursuit of knowledge and jeopardizes the quality of the degree awarded to all graduates. Any student who commits an act of scholastic dishonesty is subject to discipline. Scholastic dishonesty includes but is not limited to, cheating, plagiarism, collusion, the submission for credit of any work or materials that are attributable in whole or in part to another person, taking an examination for another person, any act designed to give an unfair advantage to a student or the attempt to commit such acts. Students who violate these rules on scholastic dishonesty are subject to disciplinary penalties, including the possibility of failure in the course/lab and/or dismissal from the university. Since dishonesty harms the individual, all students, and the integrity of the university, policies on scholastic dishonesty will be strictly and actively enforced. In addition, any sort of misbehavior or arguments with the teaching assistants will also result in serious consequences.